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Biochemistry Module, First Year Medicine

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STRUCTURES AND PROPERTIES OF SIMPLE LIPIDS

Course objectives:

-Characterize the chemical nature of the components of glycerides, cerides, and sterides.

-Characterize the chemical and enzymatic hydrolysis properties of glycerides.

-Define the physical properties of cerides.

-Define the physical properties of sterides

I/Introduction

• These are simple lipids, ternary compounds consisting of C, H, O.

• They are esters of fatty acids + alcohol.

• Three types of alcohol are esterified by fatty acids:

— Glycerol → Glycerides

— Cholesterol → Sterides

— High molecular weight alcohol → Cerides

Biological roles:

- Energy reserve

- 95-98% of dietary lipids

- in adipose tissue (reserve fats) and
- in many vegetable oils (seeds).

- Significant energy reserve in humans:

100,000 kcal

- 21% of body mass in men; 26% in women.
- Example for a 70 kg man = 11 kg in the form of triglycerides

➢ Thermal insulation and thermogenesis, for example seals and penguins have large amounts of TG stored under their skin

There are **two** types of adipose tissue:

– **white**, where TG hydrolysis ensures energy production,

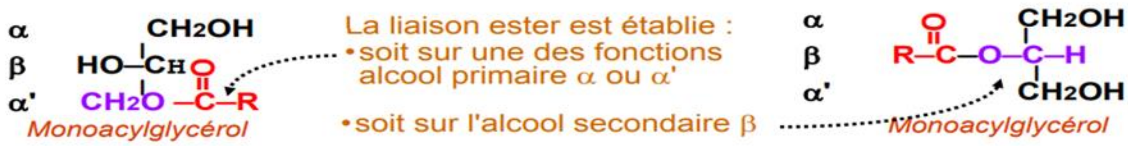
– **brown**, which ensures heat production (found in animals exposed to cold: hibernators).

II/Acylglycerols: Glycerides

A/Structure:

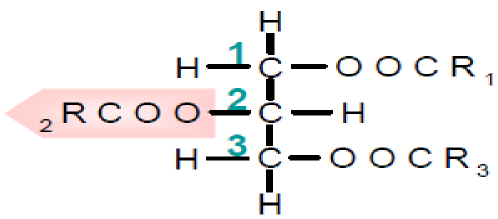
Glycerides constitute the bulk of natural lipids in terms of quantity. They are esters of glycerol and fatty acids.

- **Monoacylglycerol** (MAG): combination of **one** a fatty acid molecule with a glycerol molecule.



- **Diacylglycerol** (DAG): Diester of one glycerol molecule and **two** fatty acid molecules, which may or may not be identical.

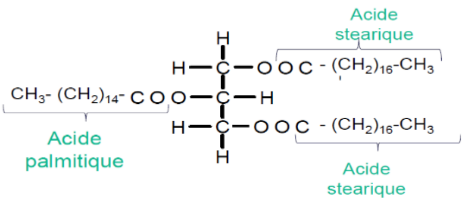
- **Triacylglycerol** (TAG) (formerly and sometimes still called triglyceride): Triester of one molecule of glycerol with **three** molecules of fatty acids, which may or may not be identical.
  In general, R1 and R3 are saturated fatty acids and R2 is preferably an unsaturated fatty acid.

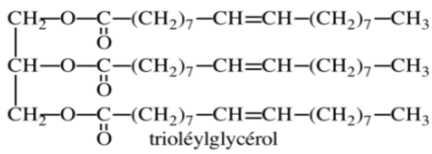


-When the fatty acid molecules that make up the di- or triester are **identical**, they are referred to as **homogeneous** diacylglycerol or triacylglycerol; otherwise, they are referred to as **mixed** diacylglycerol or triacylglycerol. Triacylglycerols are **neutral lipids**.

Exemple de triacyglycérol mixte

1,3-distéaryl-2palmityl-sn-glycérol





Trioléylglycerol=trioléine

Example of a homogeneous triacylglycerol

-Triolein is a triglyceride produced by the esterification of glycerol with three molecules of **oleic acid.**

-Tristearin is the result of the esterification of glycerol with three molecules of **stearic acid**.

B/Physicochemical properties:

1) Physical properties:

-The dominant physical property is the completely **nonpolar** nature of natural acylglycerols, mainly triacylglycerols (TAG).

Polar groups (hydroxyl or carboxyl) disappear in **ester bonds.**

- They are **insoluble** in water and highly soluble in the most nonpolar solvents such as acetone.
- When stirred in water, they form highly unstable **emulsions** that transform into a two-phase system. Surfactants, such as soaps, disperse and stabilize these **emulsions**, where TAG are suspended in the form of micelles.

2) Chemical properties

These are those of **fatty acid chains** and **esters**:

- ❖ Chemical hydrolysis

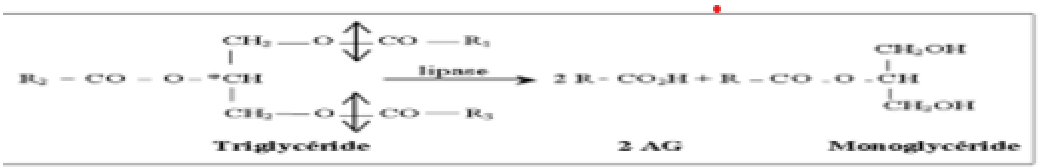
Acid treatment releases the constituents: fatty acids and glycerol.

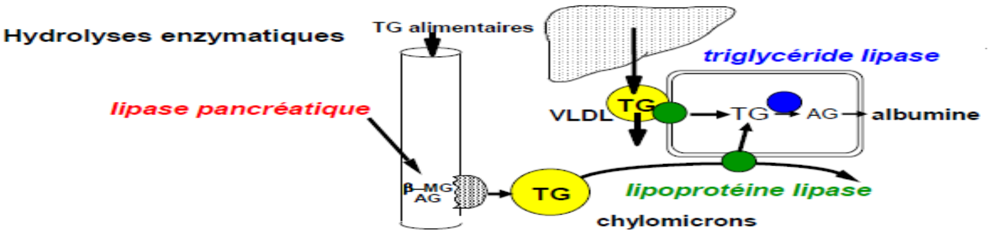
- ❖ Enzymatic hydrolysis:

Lipases hydrolyze TAG with different **specificities**.

For example, pancreatic lipase hydrolyzes them in stages in emulsion (bile salts present in the intestine) and in the presence of a protein factor called colipase.

A TAG is hydrolyzed into a diglyceride with the release of a fatty acid, and the diglyceride into 2-monoacylglycerol and a fatty acid, which are absorbed by the intestine.





❖ Saponification

Alcoholic bases **(sodium or potassium hydroxide)** heated to high temperatures **break** the **ester bonds** in glycerides, releasing fatty acids in the form of sodium salts (hard soaps) or potassium salts (soft soaps).
This reaction can also be used to deduce a saponification index.

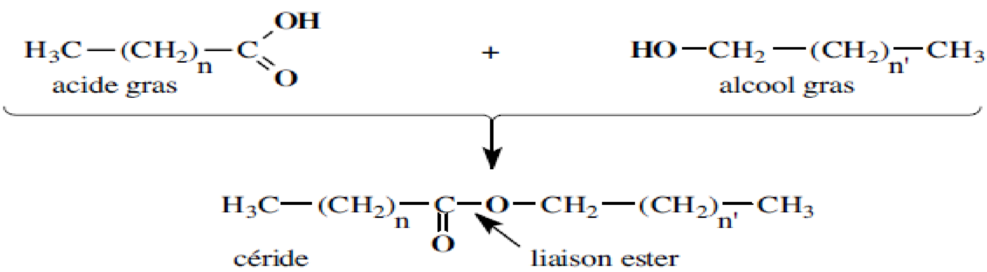
By definition, the saponification index is:

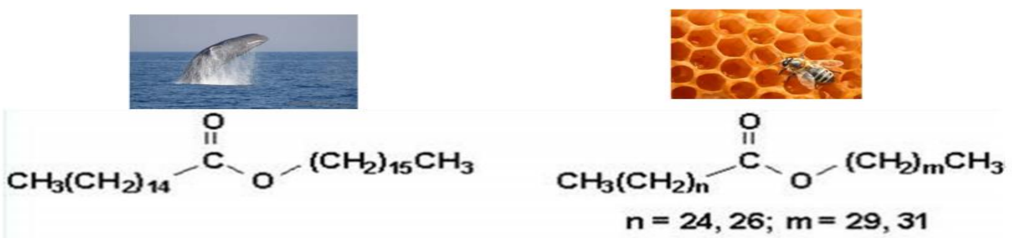
The number of **mg** of KOH required to saponify and neutralize the free acidity of **1g** of fatty substance.

III/Cerides = waxes

➢ **Structures**

-They owe their generic name to the fact that they are the main constituents of animal, vegetable, and bacterial waxes.
-Cerides are monoesters of fatty acids and long-chain aliphatic alcohols, which are generally primary alcohols with an even number of carbon atoms, saturated and unbranched.





Palmitate de cétyle
(cire du cachalot)

Cire d'abeille

- **Physicochemical properties**

-They are highly nonpolar and therefore hydrophobic, insoluble in water and soluble in solvents.
-Solid at room temperature; very high melting point.
-When saponified, they produce a long-chain alcohol and a fatty acid in the form of soap.
Here, the long-chain alcohols are hydrophobic and constitute the saponifiable fraction.

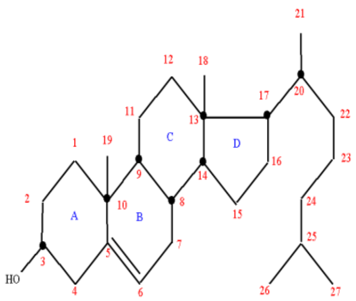
- **Role**

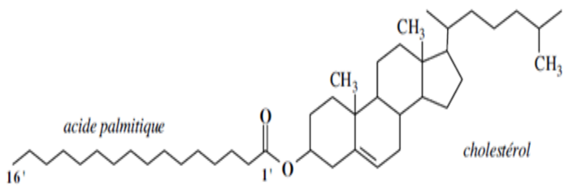
-Waxes therefore provide waterproof coatings and are good thermal insulators. For example, they cover the wings of birds, e.g., vegetable waxes.
-Use of cerides in the cosmetics industry (lotions, ointments, creams, makeup, etc.) and coatings.

IV/Sterides

- They result from the esterification of fatty acids by sterols.

*Sterols: These alcohols are derived from the steroid nucleus, a product of the condensation of four cycles in which the hydroxyl group is a secondary alcohol function always in the same position.
The most representative is cholesterol (it has a secondary alcohol function at C3 and a double bond at Δ5).





palmitate de cholestéryle

- Sterides are formed by esterification of a fatty acid on the alcohol function in position 3 of cholesterol, for example cholesterol palmitate.
- **The biological importance of cholesterol:**
  -Cholesterol is obtained from food and synthesized by the liver; it is transported in the blood in lipoproteins.
  -It is a component of cell membranes (role in fluidity).
  -Cholesterol is used in the body to synthesize three groups of molecules:
    -Steroid hormones (cortisol, testosterone, etc.)
    -Vitamin D3
    -Bile acids